■ Plot3D

`Plot3D[`*f*`, {`*x*`, `*xmin*`, `*xmax*`}, {`*y*`, `*ymin*`, `*ymax*`}]` generates a three-dimensional plot of *f* as a function of *x* and *y*.

`Plot3D[{`*f*`, `*s*`}, {`*x*`, `*xmin*`, `*xmax*`}, {`*y*`, `*ymin*`, `*ymax*`}]` generates a three-dimensional plot in which the height of the surface is specified by *f*, and the shading is specified by *s*.

`Plot3D` evaluates its arguments in a non-standard way (see page **??**). ■ The following options can be given:

`AmbientLight`	`GrayLevel[0.]`	ambient illumination level
`AspectRatio`	`1`	ratio of height to width
`BoxRatios`	`{1, 1, 0.4}`	bounding 3D box ratios
`Boxed`	`True`	whether to draw the bounding box
`ClipFill`	`Automatic`	how to draw clipped parts of the surface
`DisplayFunction`	`$DisplayFunction`	function for generating output
`Framed`	`False`	whether to draw a frame
`LightSources`	(see below)	positions and colors of light sources
`Lighting`	`False`	whether to use simulated illumination
`Mesh`	`True`	whether to draw a mesh on the surface
`Plot3Matrix`	`Automatic`	perspective transformation matrix
`PlotColor`	`True`	whether to plot in color
`PlotLabel`	`None`	a label for the plot
`PlotPoints`	`15`	the number of sample points in each direction
`PlotRange`	`Automatic`	range of values to include
`Shading`	`True`	whether to shade polygons
`ViewPoint`	`{1.3, -2.4, 2.}`	viewing position

■ `Plot3D` returns a `SurfaceGraphics` object. ■ The function *f* should give a real number for all values of *x* and *y* at which it is evaluated. There will be holes in the final surface at any values of *x* and *y* for which *f* does not yield a real number value. ■ If `Lighting->False` and no shading function *s* is specified, the surface is shaded with gray levels according to height. ■ The shading function *s* must yield values of the form `GrayLevel[`*i*`]` or `RGBColor[`*r*`, `*g*`, `*b*`]`. ■ The default light sources used are as for `Graphics3D`. ■ See page 120. ■ See also: `ListPlot3D`, `ContourPlot`, `DensityPlot`, `Graphics3D`.

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